

Emily N Peterson, PhD
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Research Interests

Methodological: Bayesian hierarchical modeling, Measurement error methods, Partial Identifiability and Recovery, Spatial disease mapping, Statistical demography and population health, Dynamics and Projections, Health surveillance.

Collaborative: Indigenous populations, Maternal health and mortality, Opioid use, Demographic component population estimation.

Educational

- University of Massachusetts Amherst, Amherst, MA. 2019. **Doctor of Philosophy** in Biostatistics. Adviser: Dr. Leontine Alkema.
- Vanderbilt University, Nashville, TN. 2015. **Master of Science in Biostatistics**. Adviser: Dr. Tatsuki Koyama.
- Penn State University, Hershey, PA. 2013. **Master of Science in Public Health Sciences**. Adviser: Dr. Vernon Chinchilli.
- Davidson College, Davidson, NC. May 2006. **Bachelor of Science in Psychology**. With a research focus in Cognitive and Neuroscience Psychology.

Appointments

- 2022-Present **Research Assistant Professor of Biostatistics**, *Emory University*.
Department of Biostatistics and Bioinformatics
Rollins School of Public Health
- 2020-2022 **Post-Doctoral Fellow in Biostatistics**, *Emory University*.
Department of Biostatistics and Bioinformatics
Rollins School of Public Health
- Advisor: Dr. Lance Waller
 - Fusing of disease mapping, small area estimation, and measurement error methodologies to assess impact of spatial patterns in population uncertainty (uncertainty in local sample sizes) and covariate and confounder uncertainty.
- 2015-2019 **Graduate Research Assistant**, *University of Massachusetts Amherst*.
Department of Biostatistics and Epidemiology
School of Public Health & Health Sciences
- Advisor: Dr. Leontine Alkema
 - Bayesian hierarchical misclassification models to assess national trends of maternal mortality in data sparse contexts.

Research Support

PENDING

- 2026-2030 *Improving Statistical Methods for Small Area Estimates of Public Health Indicators and Demographic Characteristics*
- Funding: NICHD- R01HD092580
 - Role: Co-I

ONGOING

- 2025-2028 *Data Integration Methods to Improve Small-Area Mortality Estimation for Tribal and Rural Populations*
- Funding: NIMHD- R01MD020144-01A1

- 2026-2028 **Role: PI**
Bayesian Reconstruction of Under-Five Small Area Populations: Correcting for Undercount Bias and Source Uncertainty
 - Funding: NICHD- R21
- 2026-2028 **Role: PI**
A Bayesian Approach to Assess Measurement Error in the Pregnancy Mortality Surveillance System
 - Funding: Independent Contract
 - Role: Sub-Award Principal Investigator
- 2024-2026 *Development of an Advanced Data Science Workflow for Culturally Relevant Surveillance of Gestational Diabetes in Tribal Community and Associative Social Determinants of Health*
 - Funding: Rollins School of Public Health Dean's Pilot Award
 - **Role: PI**
- 2023-2025 *Substance Use in Pregnancy and the Morbidity Mortality Risk Environment (PROMISE)*
 - Funding: NIH-NIDA R01DA059182-01A1
 - Role: Co-I
- 2023-2025 *The SLEEP study: Sleep Effects on Post-stroke Rehabilitation*
 - Funding: NIH-NINR 5R01NR018979-04 (PI Klingman)
 - Role: Statistical Support
- 2023-2027 *HERCULES: Exposome Research Center*
 - Funding: NIH-NIEHS 5P30ES019776-12 (PI Marsit)
 - Role: Statistical Support
- 2023-2030 U54 Maternal Health Research Center for Excellence
 - Funding: NIH-NIDA 1U54HD113292-01 (Cooper, Hernandez)
 - Role: Statistical Support

COMPLETED

- 2023-2025 *Limited interaction cohort to identify determinants of viral suppression in MSM and transfeminine individuals living with HIV (D-LITE)*
 - Funding NIH-NIEHS 1UG3AI176853-01 (PI Sullivan, Siegler)
 - Role: Statistical Support
- 2023-2030 *Center to Advance Reproductive Justice and Behavioral Health among Black Pregnant/Postpartum Women and Birthing People (CORAL)*
 - Funding NIH-NICHD 1U54HD113292-01 (PI Hernandez, Cooper, Dunlop)
 - Role: Statistical Support
- 2018-2023 *Spatial Uncertainty in Small Area Population Inference from Survey & Administrative Data*
 - Funding NIH-NICHHD R01HD092580 (PI Waller)
 - Post-Doctoral Fellow & Co-I
- 2016-2020 *A Bayesian approach to estimate maternal mortality using national civil registration vital statistics data accounting for reporting errors*
 - Funding: Independent Contract
 - Role: Sub-Award Principal Investigator
- 2024-2024 *A data fusion approach to assess small area opioid use disorders rates in data sparse contexts*
 - Funding sub-contract PJ 66368 (PI Yarborough)
 - Role: Statistical Consult

Honors

2025-2026 ASA StatsForward Fellowship, American Statistical Association — *National early-career fellowship recognizing leadership potential and contributions to statistics and data science*

2026 Invited Reviewer, NIH CSR PDAH Study Section — Selected to provide scientific peer review of NIH research grant applications.

Publications

* *First authored by student or post-doc for who I was their primary mentor.*

1. **EN Peterson**, Q Wang. *FC-SPIDR: A Two-Step Recoverability Diagnostic. Bayesian Analysis*. Submitted. 2026.
2. **EN Peterson**, P Gutman, B Garrett, K Komro, LA Waller, M Kramer. Bayesian Spatial Data Fusion for Estimating Disease Prevalence under Measurement Error and Incomplete Coverage: An Application to American Indian and Alaska Native Gestational Diabetes. *Journal of the Royal Statistical Society: Series C*. Submitted. 2026.
3. **EN Peterson**, A Manatunga, KJ Klingman, GD Fulk. A Bayesian approach to modeling joint trends and dependencies in irregular multivariate time series. *Statistics in Medicine*. Submitted. 2026.
4. **EN Peterson**, A Edwards, M Wetzel, LA Waller, H Cooper, C Yarbrough. A Bayesian Spatio-Temporal Top-Down Framework for Opioid Use Disorder Risk Under Data Sparsity. *Bayesian Analysis*. Submitted. 2026. Available on arXiv: <https://arxiv.org/abs/2506.02303>
5. *K Schrote, M Kramer, **EN Peterson**. A Bayesian Hierarchical Spatial Mapping Approach to Assess Gestational Diabetes Mellitus Risk Among Immigrant Populations in Georgia. *American Journal of Epidemiology*. Submitted. 2026.
6. *C Peterkin, LA Waller, **EN Peterson**. A Bayesian spatio-temporal approach to assess small area trends in COVID-19 related excess opioid mortality among drug users in Georgia. *Journal of the Royal Statistical Society: Series A*. Submitted. 2026. Available on arXiv: <https://arxiv.org/abs/2503.07918>
7. **EN Peterson**, RC Nethery, JT Chen, N Krieger, PD Waterman, BA Coull, LA Waller. A Bayesian Spatial Berkson error approach to estimate small area opioid mortality rates accounting for population-at-risk uncertainty. *Spatial and Spatio-Temporal Epidemiology*. Accepted. 2025.
8. **EN Peterson**, G Guranich, JA Cresswell, L Alkema. A Bayesian approach to estimate maternal mortality globally using national civil registration vital statistics data accounting for reporting errors. *Statistics and Public Policy*. 2024; 11(1):2286313.
9. **EN Peterson**, RC Nethery, T Padellini, JT Chen, BA Coull, FB Piel, LA Waller. A Bayesian hierarchical small area population model accounting for data source specific methodologies from American Community Survey, Population Estimates Program, and Decennial census data. *The Annals of Applied Statistics*. 2024; 18(2):1565–1595.
10. JS Varghese, **EN Peterson**, MK Ali, N Tandon. Advancing diabetes surveillance ecosystems: a case study of India. *The Lancet Diabetes & Endocrinology*. 2024.
11. GD Fulk, **EN Peterson**, KJ Klingman. Abstract TP93: A Bayesian Hierarchical Time Series Model to Forecast Patient Recovery Outcomes After an Acute Stroke When Activity Monitor Data is Incomplete. *Stroke*. 2024; 55(Suppl 1):ATP93.
12. Y Li, BA Coull, N Krieger, **EN Peterson**, LA Waller, JT Chen, RC Nethery. Impacts of census differential privacy for small-area disease mapping to monitor health inequities. *Science Advances*. 2023; 9(33):eade8888.
13. JA Kempker, E Stearns, **EN Peterson**, LA Waller. US Adult Critical Care Beds Per Capita: A 2021 County-Level Cross-Sectional Study. *Critical Care Explorations*. 2023; 5(3):e0868.
14. **EN Peterson**, AB Moller, A Gemmill, D Chou, L Alkema. A Bayesian temporal hierarchical model to assess levels of misclassification error in national vital registration maternal mortality data. *Statistics in Medicine*. 2022; 41(14):2483–2496.
15. RC Nethery, JT Chen, N Krieger, PD Waterman, **EN Peterson**, LA Waller, BA Coull. Statistical implications of endogeneity induced by residential segregation in small-area modeling of health inequities. *The American Statistician*. 2022; 76(2):142–151.
16. NB Jain, JE Kuhn, GD Ayers, A Song, **EN Peterson**. Geographical Variation in Rates of Shoulder and Knee Arthroscopy in U.S. States and Relationship to Orthopedist Density in Surgeon Volume. *JAMA Network Open*. 2019; 2(12). doi: 10.1001/jamanetworkopen.2019.17315.
17. NB Jain, GD Ayers, **EN Peterson**, MB Harris, L Morse, KC O'Connor, E Garshick. Traumatic spinal cord injury in the United States, 1993–2012. *JAMA*. 2015; 313(22):2236–2243.
18. A Morandi, LM Solberg, R Habermann, P Cleeton, **EN Peterson**, EW Ely, J Schnelle. Documentation and Management of Words Associated with Delirium Among Elderly Patients in Postacute Care: A Pilot Investigation. *JAMDA*. 2009; 34–339.

19. SF Simmons, **EN Peterson**, C You. The Accuracy of Monthly Weight Assessments in Nursing Homes: Implications for the Identification of Weight Loss. *Journal of Nutrition, Health and Aging*. 2009; 13(3):284–288.
20. JF Schnelle, SF Simmons, L Beuscher, **EN Peterson**, R Habermann, F Leung. Prevalence of Constipation Symptoms in Nursing Home Residents. *Journal of Gerontology*. 2009.

Guest Lectures and Invited Conference presentations

- Spatial Statistics 2025: “A Bayesian spatial reconstruction of the Net-Migrant population.” Oral Presentation.
- United States Census Bureau 2025: “A Bayesian hierarchical small area population model accounting for data source specific methodologies from U.S. Census data.” Invited talk.
- ISBA 2024: “A Bayesian hierarchical small area population model accounting for data source specific methodologies from American Community Survey, Population Estimates Program, and Decennial Census Data”
- SAE 2024: “Incorporating heterogeneous types of uncertainty in small area estimates from multiple demographic data sources”
- PAA 2023 Session Chair to the Bayesian Methods Section.
- PAA 2023: “A Bayesian Spatial Berkson Error approach to estimate small area disease rates accounting for population-at-risk uncertainty.” Oral presentation to be given to the “Incorporating Concepts of Space into Mortality Analyses” Section.
- Southern Regional Council on Statistics 2022: “Bayesian methods for the assessment of measurement errors with the use of multiple data sources with application to small area estimation.”
- PAA 2022: “A Bayesian hierarchical small area population model accounting for data source specific methodologies from U.S. Census data.” Oral presentation given to the “Statistical Methods for Measurement Error and Small Area Estimation” Section.
- Maynooth University Biostatistics Seminars: “A Bayesian hierarchical model framework to estimate measurement error in data sparse settings: Application in maternal mortality and U.S. Census data.”
- University of Massachusetts Bayesian Working Group: “A Bayesian hierarchical small area population model accounting for data source specific methodologies from U.S. Census data.”
- University of Massachusetts Bayesian Working Group: “Incorporating population-at-risk uncertainty into spatial disease mapping models.”
- NESS 2019 Student Paper Competition: “A Bayesian temporal hierarchical model for the assessment of misclassification errors in maternal mortality count data: Application in sparse data settings”
- JSM 2018 Poster: “Counting Maternal Deaths? You Better Bayes It! A systematic assessment of underreporting and misclassification in registration of maternal deaths in high- and middle-income countries using a Bayesian modeling approach”
- PAA 2018 Poster: “Counting Maternal Deaths? You Better Bayes It! A systematic assessment of underreporting and misclassification in registration of maternal deaths in high- and middle-income countries using a Bayesian modeling approach”
- NESS 2018 Poster: A Bayesian approach to modeling CRVS reporting errors of maternal mortality”
- UMASS SPH Environmental Health Science research day Poster: “Counting Maternal Deaths? You Better Bayes It! A Bayesian Approach to modeling incorrect reporting of maternal deaths”
- World Health Organization Maternal Mortality Interagency Group (WHO-MMEIG). Report on estimation of reporting errors of vital registration (VR) maternal mortality data.

Academic Instruction

- “Modern Regression Methods”. Emory University. BIOS 526. Fall 2024, 2025. [4 credits]
Course evaluations: Fall 2024: 4.47/5 overall, 4.66/5 inclusive course environment; Fall 2025: 4.66/5 overall, 4.76/5 inclusive course environment.
- “Advanced Geographic Information Systems”. Emory University. DATA 532. Spring 2024. [2 credits]
Course evaluations: Fall 2024: 4.33/5 About the Course, 4.63/5 Inclusive Course Environment.
- “Advanced Geographic Information Systems”. Emory University. INFO 532. Fall 2023. [2 credits]
- “Introduction to Geographic Information Systems”. Emory University. INFO 530. Spring 2023. [2 credits]
- “Advanced Geographic Information Systems”. Emory University. INFO 532. Spring 2023. [2 credits]
- “Introduction to Geographic Information Systems”. Emory University. INFO 530. Fall 2021. [2 credits]

- “Advanced Geographic Information Systems”. Emory University. INFO 532. Fall 2021. [2 credits]
- “Introduction to Bayesian Statistics”. University of Massachusetts. BIOS 697. Spring 2017. [3 credits]
- “Introduction to Biostatistics”. University of Massachusetts. BIOSTATS 540. Spring 2016. [3 credits]

Faculty Advisor

- Dissertation Advisory Committee
 - Delante Moore, PhD 2025
 - Michael Dorfler, 2026
- Thesis/ Capstone Advisor
 - Julianna Chapman, MSPH 2025
 - Kaitlin Schrote, MSPH 2025- Nominee for RSPH Shepherd’s Thesis Award
 - Yujia Dou, MSPH 2025
 - Cyen Peterkin, MSPH 2024- Nominee for RSPH Shepherd’s Thesis Award
 - Yixiang Hong, MSPH 2024- Winner of BIOS Shepherd’s Capstone Award
 - Sarah Savariyar, MSPH 2024
 - Wenchao Ma, MSPH 2024

Professional Experience

Research Assistant Professor- Department of Biostatistics, Emory University, Atlanta, GA. Jul 2022-Present

- § Develop an independent research program in **Bayesian methodology for incomplete and imperfect data**, with emphasis on measurement error, partial identifiability and recoverability, spatial and small-area estimation, and integration of multiple data sources.
- § **Principal Investigator of an NIH-funded R01** developing Bayesian data-integration and measurement-error methods to improve small-area mortality estimation for Tribal and rural populations.
- § **Principal Investigator of an NIH R21** developing Bayesian methods to reconstruct annual county-level populations of children under age five by integrating multiple Census Bureau data products while accounting for data-source-specific uncertainty and measurement error.
- § Develop new Bayesian statistical methodology for **partial identifiability, measurement error, and data-source uncertainty**, including diagnostic approaches for assessing parameter recoverability and spatial models that propagate uncertainty in population-at-risk estimates.
- § Lead and collaborate on the development of **Bayesian spatial and small-area methods for population health**, with applications to opioid use disorder and mortality, maternal and perinatal health, Indigenous population health, and demographic estimation.
- § Serve as a statistical methodologist with the **CDC Maternal Mortality Prevention Team**, developing Bayesian approaches to characterize and account for measurement and misclassification error in maternal mortality surveillance.
- § Collaborate with investigators at Emory, Harvard T.H. Chan School of Public Health, Mercer University, Cherokee Nation Public Health, the UK Small Area Health Statistics Unit, and other institutions on methodological and population-health research.
- § Mentor graduate students and junior researchers in Bayesian, spatial, and population-health methodology, including supervision of student-led methodological and applied research projects.

Post-Doctoral Fellow- Department of Biostatistics, Emory University, Atlanta, GA. Feb 2020-Jun 2022

- Developed a Bayesian hierarchical small area misclassification model to estimate extent of misclassification errors of maternal mortality by U.S. Pregnancy Mortality Surveillance Systems. In collaboration with CDC Maternal Mortality Surveillance Group and Michael Kramer at Emory University Epidemiology Department.
- Developed Bayesian methods to estimate reporting errors in the reporting of age-cohort specific adult survival probabilities in sibling survival histories using a two stage approach. In collaboration with

University of Massachusetts Amherst and Stephane Helleringer at New York University Abu Dhabi Social Research and Public Policy Department.

- Developed Bayesian spatial-temporal methods to estimate population size of blue whales using multiple data sources, accounting for data source specific biases. In collaboration with Oregon State Marine Mammal Institute.
- Developed a Bayesian hierarchical small area population model to estimate true population size accounting for data source specific methodologies in U.S. Census data. In collaboration with Harvard T.H.Chan School of Public Health, and UK Small Area Health Statistics Unit (corresponding publication listed).
- Developed a Bayesian approach to estimate maternal mortality using national civil registration vital statistics data accounting for reporting errors. In collaboration with the World Health Organization.

Research Assistant - Department of Biostatistics, University of Massachusetts Amherst, Amherst, MA. Sep 2015-Dec 2019.

- Developed a Bayesian hierarchical model to estimate the extent of reporting errors of maternal mortality by national vital registration systems. (Corresponding publication listed)
- Developed and implemented a data extraction framework for a large-scale systematic search of scientific literature related to misclassification of maternal deaths within countries, which is currently used by the WHO Maternal Mortality Interagency Group.
- Developed R packages to automate the data collection, cleaning, and compilation processes of maternal mortality data from article reviews, and WHO published surveys.
- Completed coursework in the following:
 - Bayesian Methodology and Theory
 - Machine Learning
 - Modern Applied Methods
 - Probability Theory and Statistical Inference
 - Longitudinal and Mixed Models
 - Time Series Analysis
 - Causal Inference
 - Spatio-Temporal Modeling
 - Survival Analysis
 - Clinical Trials
 - Population Research and Demography
 - Advanced R
- Completed workshops
 - STAN NESS 2018 led by Michael Betancourt

Research Associate- Department of Biostatistics, Vanderbilt University Medical Center, Nashville, TN. Jan 2014-Aug 2015.

- Data Analyst for Physical Medicine and Rehabilitation of Rotator Cuff Tears in older populations
- Performed exploratory and regression analyses to compare surgical and non-surgical treatment outcomes of rotator cuff tears
- Developed methods on the use of propensity scores with over-parametrized regression models
- Worked as data analyst and research assistant for ROW Outcomes Study
- Managed Vanderbilt REDcap data collection for rotator cuff tear surveys
- Created mock data for testing error checks.
- Prepared statistical analysis plan for subsequent studies on treatment efficacy.

Independent Contracts

- Consultant/Statistical Methodologist, CDC Maternal Mortality Prevention Team 2023-2024, 2026-present
 - Develop Bayesian methods to assess and account for measurement and misclassification error in the Pregnancy Mortality Surveillance System.

- Provide statistical expertise on maternal mortality surveillance, with emphasis on quantifying uncertainty arising from imperfect classification and reporting.
- Consulted on work sponsored by the WHO on maternal mortality misclassification methods. Dec 2020-current.
 - Provide expert input on specifications for data inputs for the maternal mortality database of specialized studies to ensure it aligns and functions well with the R code package.
 - Publication of journal article that makes substantive contribution toward improving misclassification of maternal mortality within vital statistics systems.
- Consulted on work sponsored by the WHO on global maternal mortality estimation. Apr -Sept 2016-2019.
 - Developed an R package for maternal mortality specialized study data compilation.
 - Developed a framework for data collection and compilation of studies reporting on national levels of maternal mortality.
 - Reported and presented statistical methods for measuring data quality metrics of vital statistics reporting of maternal mortality.

Committee Memberships

- Member of the Faculty Advisory Council at RSPH 2025-2028.
- Member of the RSPH Maternal Mortality and Morbidity Working Group, Rollins School of Public Health, Emory University 2022-Present
- Member of the PhD Admission Committee, Department of Biostatistics and Bioinformatics, Rollins School of Public Health, Emory University 2022-Present
- Member of the Biostatistics Diversity, Equity, and Inclusion Committee, Department of Biostatistics and Bioinformatics, Rollins School of Public Health, Emory University 2022-Present
- Member of Rollins School of Public Health Diversity, Equity, and Inclusion Committee, Rollins School of Public Health, Emory University 2023-Present

Memberships

- 2016-Present Member of Population Association of America
- 2015-Present Member of the American Statistical Association
- 2017-Present Member Society of Epidemiologic Research
- 2018-Present Member of ENAR

Reviewer Work

- Annals of Epidemiology
- Journal of Survey Statistics and Methodology
- Population of Health Metrics
- Ecology
- Epidemiology
- Spatial and Spatio-Temporal Epidemiology
- Annals of Applied Statistics
- The American Statistical Association
- Journal of the Royal Statistical Society Series A

Research Leadership & Professional Activities



- **Founder and Director, LENS (Learning, Estimation, and Nonidentifiability in Statistics), Emory University. 2026–Present.** Founded an Emory-hosted, cross-institutional research community bringing together statisticians working on identifiability and recoverability, measurement error, small-area estimation, hierarchical and spatial modeling, and data integration.
- Leader of the Interdisciplinary Bayesian Spatial Working Group, Emory University
- Member of weekly Spatial statistics working group with Lance Waller's Lab, Emory University
- Member of weekly Bayesian Statistics Working Group with Leontine Alkema's Lab at University of Massachusetts Amherst
- Member of weekly Statistics reading club at Vanderbilt University